

Chapter 5

Potential Energy and Energy Conservation

Introduction

- How do energy concepts apply to the descending sandhill crane?
- We will see that we can think of energy as being stored and *transformed* from one form to another.



Gravitational potential energy

- When a particle is in the gravitational field of the earth, there is a gravitational potential energy associated with the particle:

$$U_{\text{grav}} = mgy$$

Gravitational potential energy associated with a particle

Vertical coordinate of particle (y increases if particle moves upward)

Mass of particle

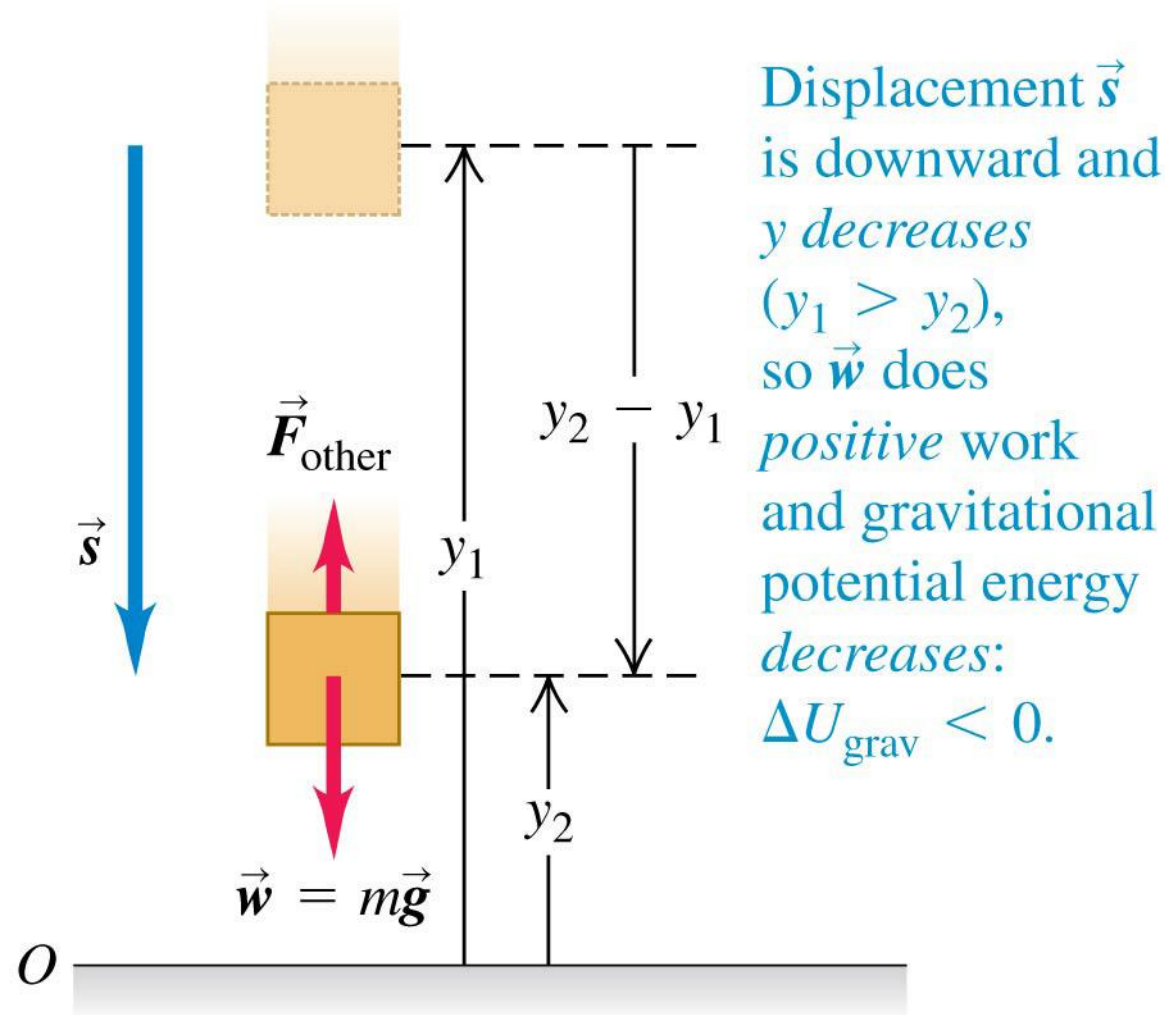
Acceleration due to gravity



- As the basketball descends, gravitational potential energy is converted to kinetic energy and the basketball's speed increases.

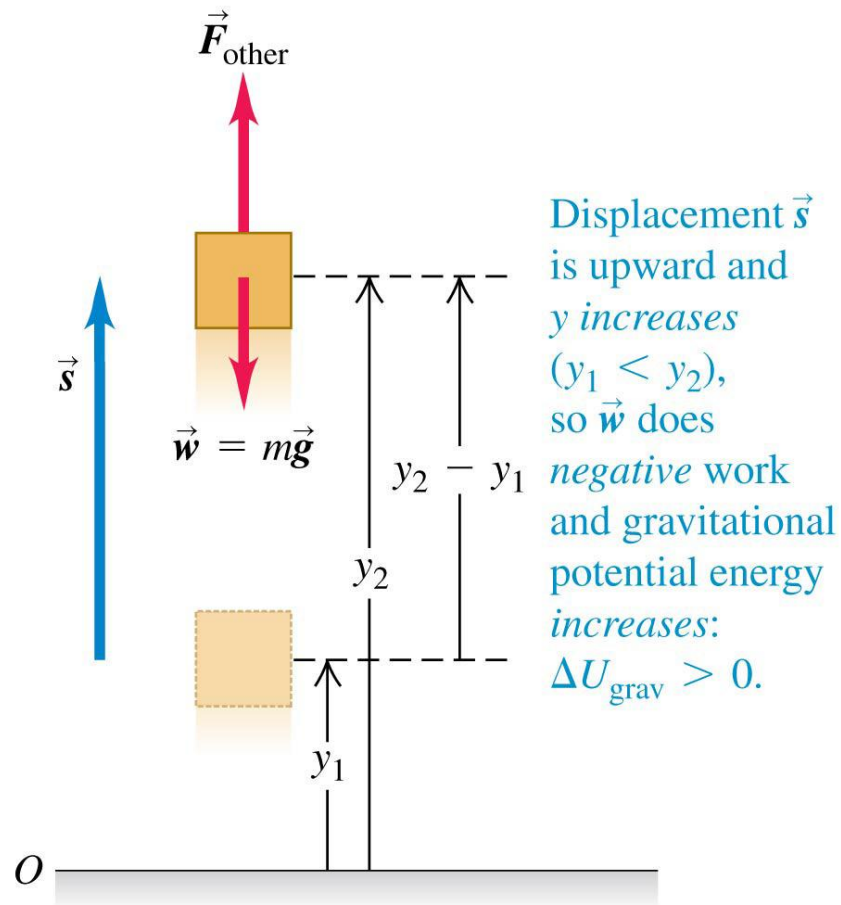
Gravitational potential energy

- The change in gravitational potential energy is related to the work done by gravity.



Gravitational potential energy

- When the body moves up, y increases, the work done by the gravitational force is negative, and the gravitational potential energy increases.



The conservation of mechanical energy

- The **total mechanical energy** of a system is **the sum of its kinetic energy and potential energy**.
- A quantity, under the change of the system, that always has the same value over time is called a **conserved quantity**.
- When **only** the force of gravity does work on a system, the total mechanical energy of that system is conserved.
- This is an example of the **conservation of mechanical energy**.

If only the gravitational force does work, total mechanical energy is conserved:

Initial kinetic energy

$$K_1 = \frac{1}{2}mv_1^2$$

Initial gravitational potential energy

$$U_{\text{grav},1} = mgy_1$$

$$K_1 + U_{\text{grav},1} = K_2 + U_{\text{grav},2}$$

Final kinetic energy

$$K_2 = \frac{1}{2}mv_2^2$$

Final gravitational potential energy

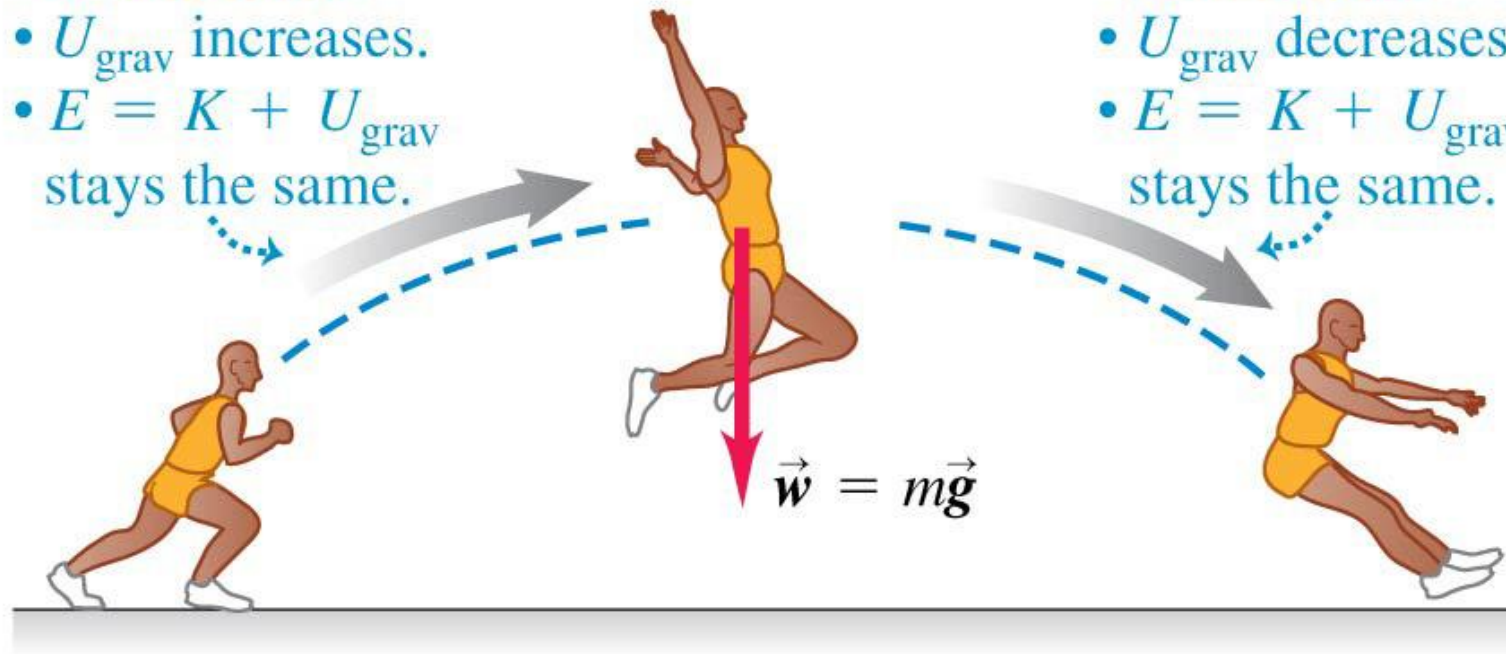
$$U_{\text{grav},2} = mgy_2$$

The conservation of mechanical energy

- When only the force of gravity does work on a system, the total mechanical energy of that system is conserved.

Moving up:

- K decreases.
- U_{grav} increases.
- $E = K + U_{\text{grav}}$ stays the same.



Moving down:

- K increases.
- U_{grav} decreases.
- $E = K + U_{\text{grav}}$ stays the same.

When forces other than gravity do work

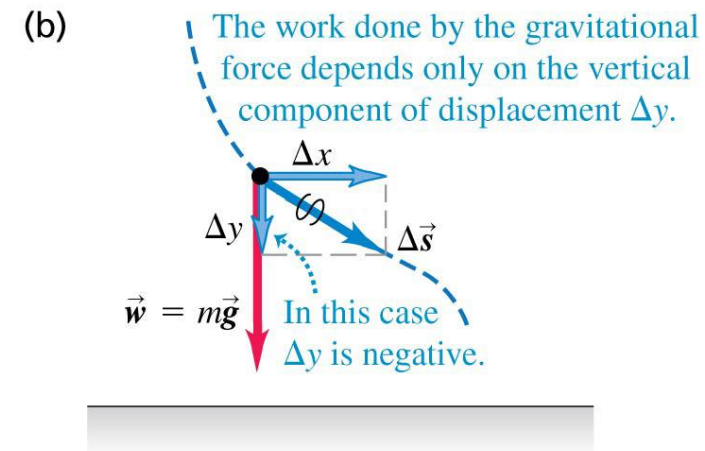
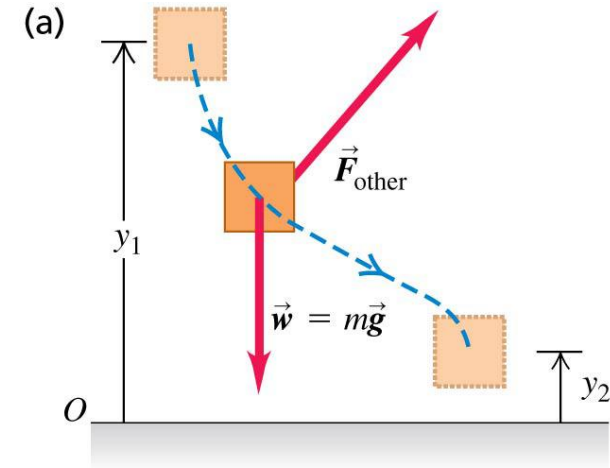


- $\vec{F}_{\text{other}}$ and $\vec{s}$ are opposite, so $W_{\text{other}} < 0$.
- Hence $E = K + U_{\text{grav}}$ must decrease.
- The parachutist's speed remains constant, so K is constant.
- The parachutist descends, so U_{grav} decreases.

Work and energy along a curved path

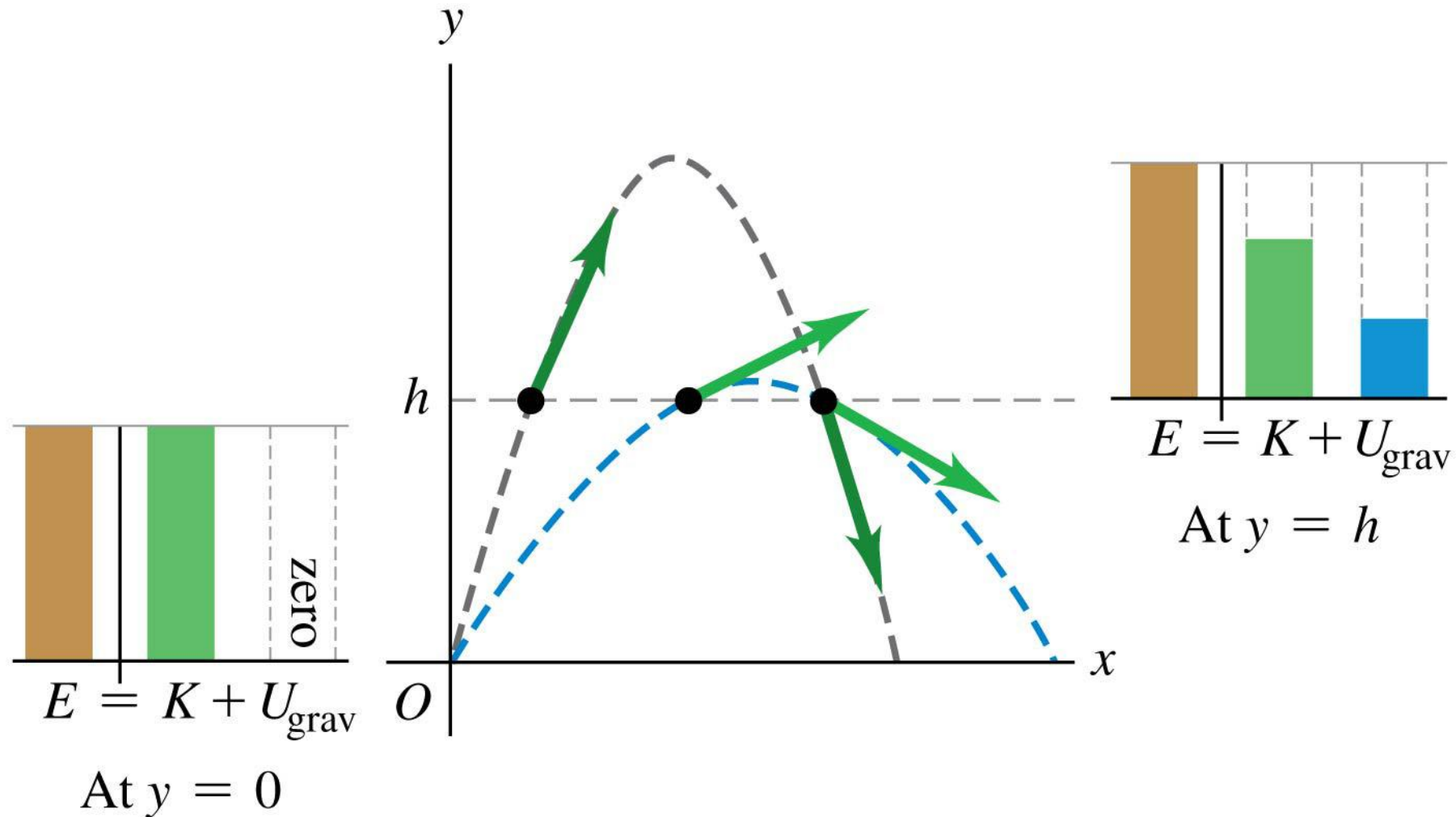
- We can use the same expression for gravitational potential energy whether the body's path is curved or straight.

$$\begin{aligned} W_{\text{grav}} &= \int_{\vec{r}_1}^{\vec{r}_2} (-mg\hat{y}) \cdot d\vec{r} \\ &= - \int_{y_1}^{y_2} mg \, dy \\ &= mgy_1 - mgy_2 \end{aligned}$$

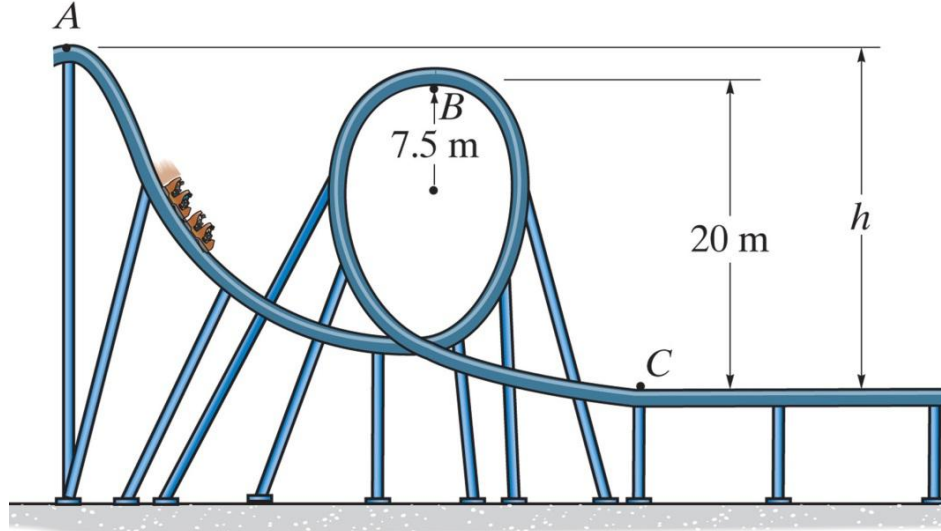


Conceptual Example

- Two identical balls leave from the same height with the same speed but at different angles.



Example



Given: The 800-kg roller coaster car is released from rest at A.

Find: The minimum height, h , of Point A so that the car travels around inside loop at B without leaving the track. Also find the velocity of the car at C for this height, h , of A.

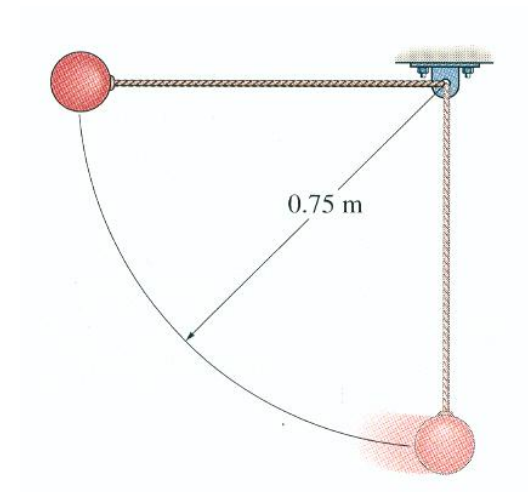
Quiz

1. The principle of conservation of energy is usually _____ to apply than the principle of work & energy.

- A) harder B) easier
C) the same amount of work D) It is a mystery!

2. If the pendulum is released from the horizontal position, the velocity of its bob in the vertical position is _____.

- A) 3.8 m/s B) 6.9 m/s
C) 14.7 m/s D) 21 m/s



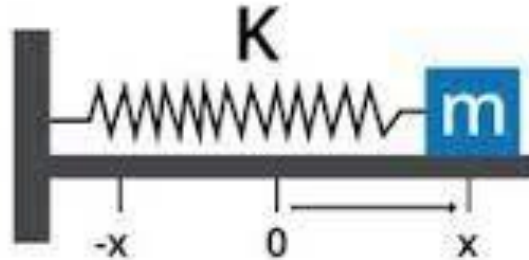
Elastic potential energy

- A body is **elastic** if it returns to its original shape after being deformed.
- **Elastic potential energy** is the energy stored in an elastic body, such as a spring:

Elastic potential energy stored in a spring $\rightarrow U_{\text{el}} = \frac{1}{2} k x^2$

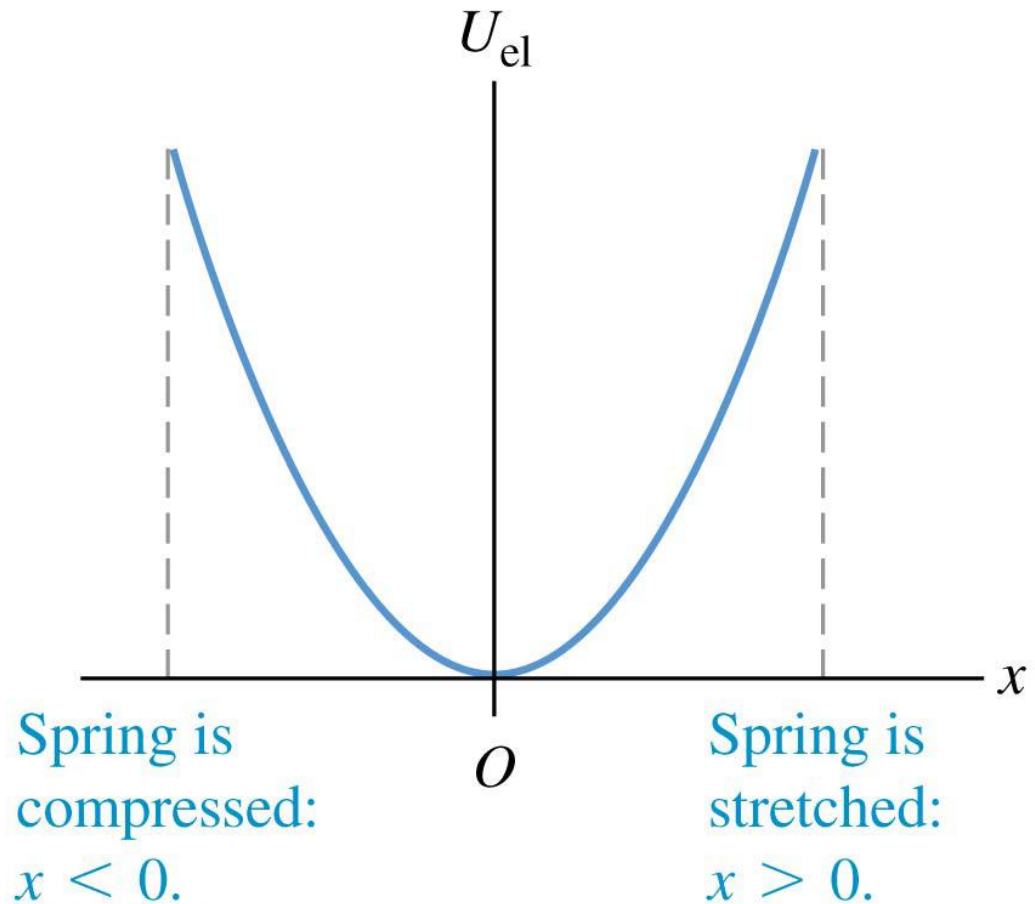
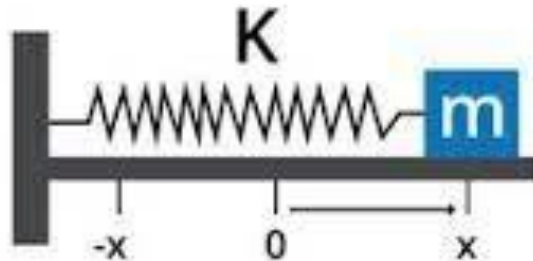
Force constant of spring k

Elongation of spring
($x > 0$ if stretched,
 $x < 0$ if compressed)



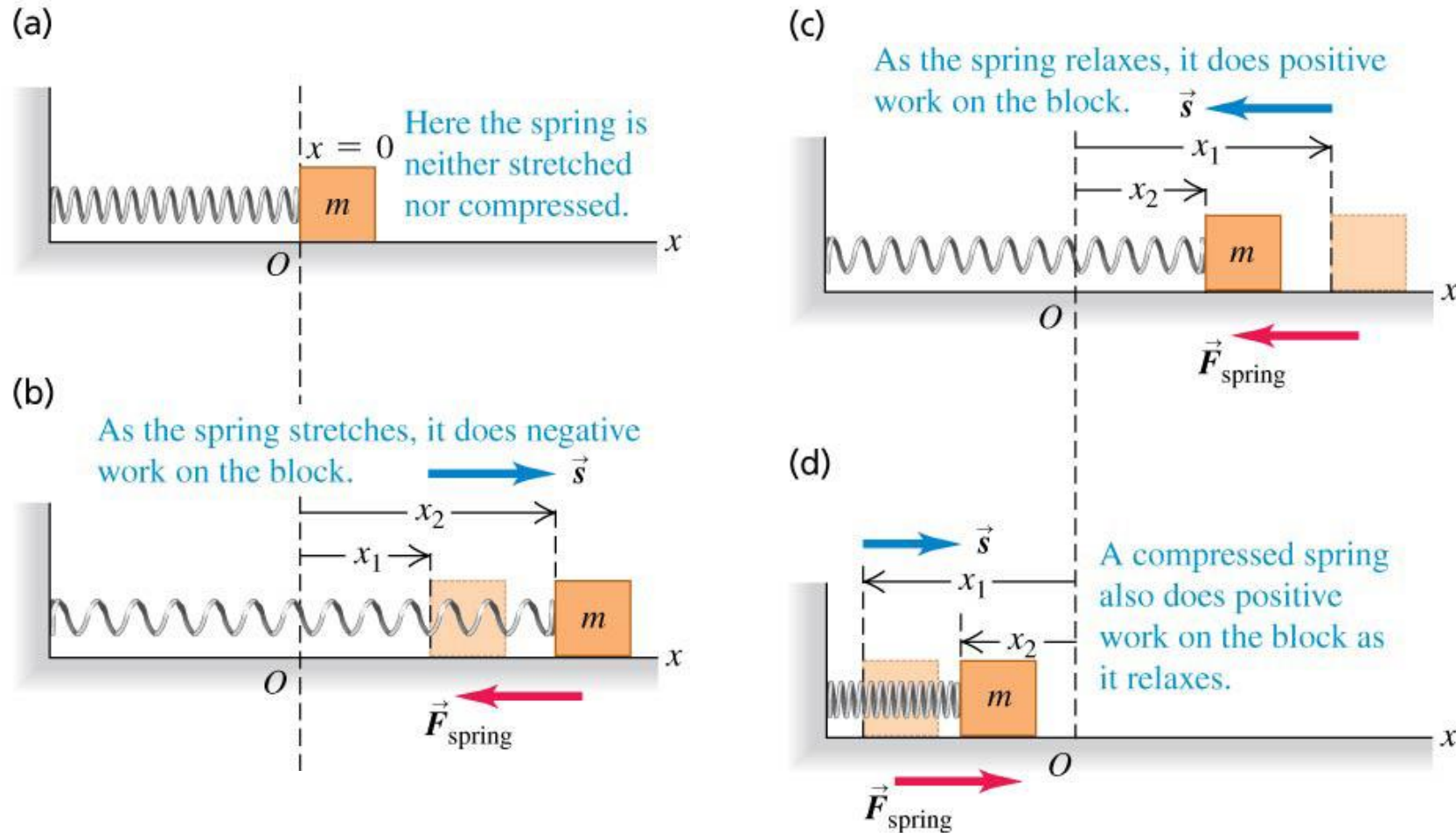
Elastic potential energy

- The graph of elastic potential energy for an ideal spring is a parabola.
- x is the extension or compression of the spring.
- Elastic potential energy is never negative.



Work done by a spring

- The figure below shows how a spring does work on a block as it is stretched and compressed.



Conservation of energy

Sum of kinetic energy and elastic potential energy remains constant.

Initial kinetic energy
 $K_1 = \frac{1}{2}mv_1^2$

Initial elastic potential energy
 $U_{el,1} = \frac{1}{2}kx_1^2$

Final kinetic energy
 $K_2 = \frac{1}{2}mv_2^2$

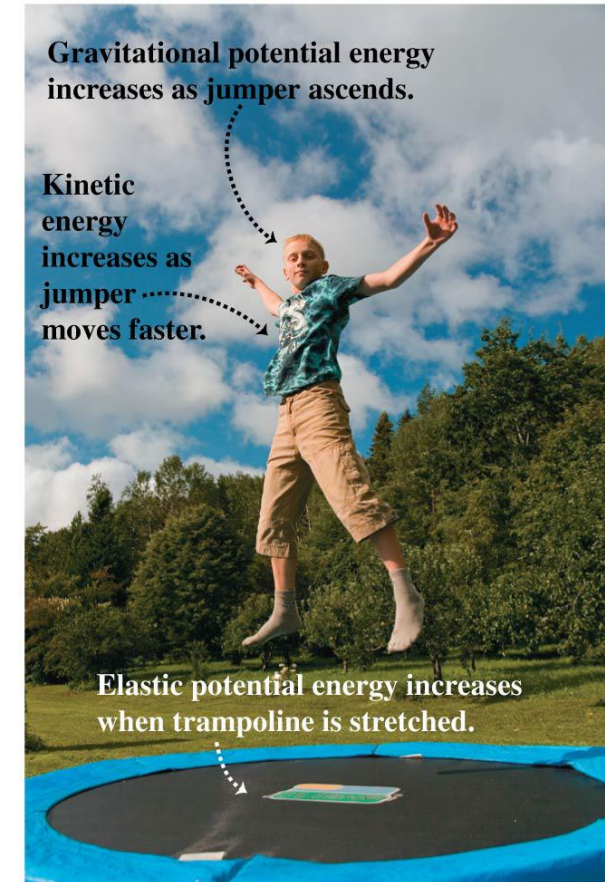
Final elastic potential energy
 $U_{el,2} = \frac{1}{2}kx_2^2$

$$K_1 + U_{el,1} = K_2 + U_{el,2}$$

Situations with both gravitational and elastic forces

- When a situation involves both gravitational and elastic forces, the total potential energy is the *sum* of the gravitational potential energy and the elastic potential energy:

$$U = U_{\text{grav}} + U_{\text{el}}$$



The conservation of mechanical energy

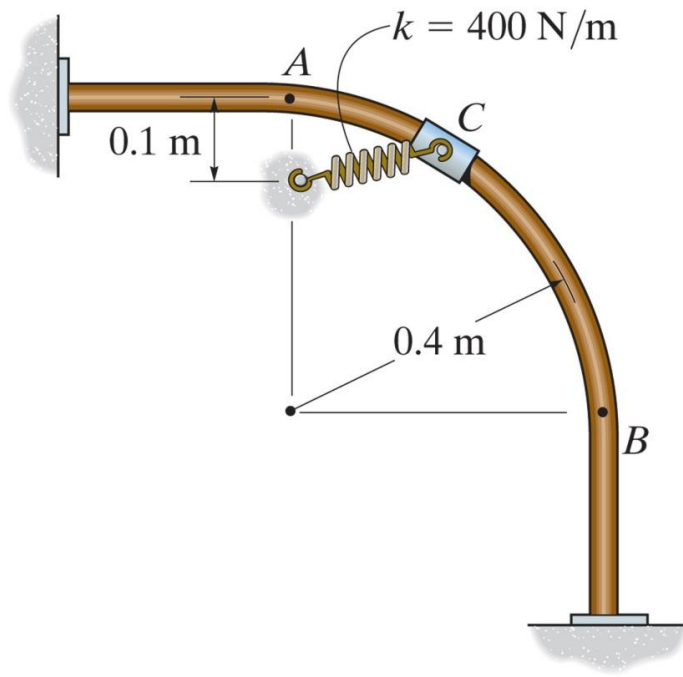
- The total mechanical energy of a system is the sum of its kinetic energy, gravitational potential energy, and elastic potential energy

$$K_1 + U_{\text{grav},1} + U_{\text{el},1} = K_2 + U_{\text{grav},2} + U_{\text{el},2}$$

Quiz

1. The potential energy of a spring is _____
A) always negative. B) always positive.
C) positive or negative. D) equal to kx .
2. When the potential energy of a conservative system increases, the kinetic energy _____
A) always decreases. B) always increases.
C) could decrease or increase. D) does not change.

Example

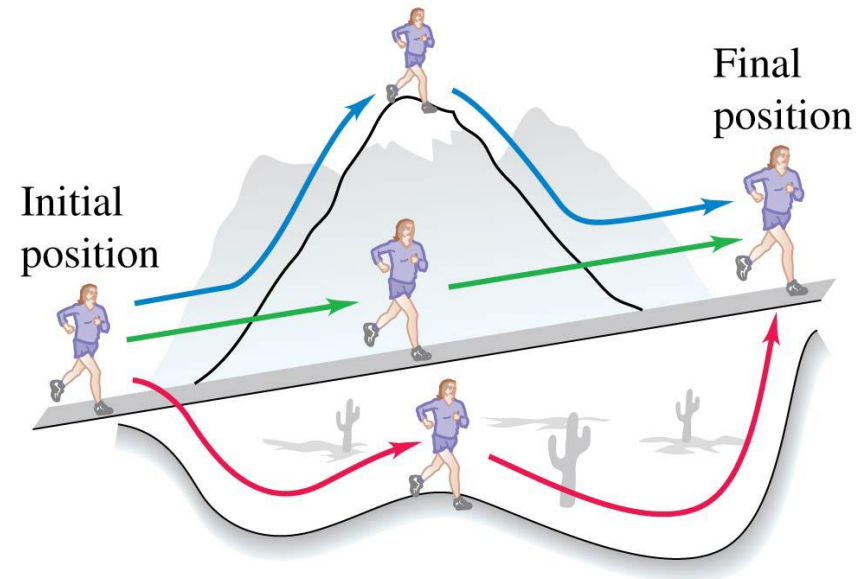


Given: The 4-kg collar, C, has a velocity of 2 m/s at A. The spring constant is 400 N/m. The unstretched length of the spring is 0.2 m.

Find: The velocity of the collar at B.

Conservative and nonconservative forces

- A **conservative force** allows conversion between kinetic and potential energy.
- Gravity and the spring force are conservative.
- The work done by a conservative force such as gravity depends on only the endpoints of a path, not the specific path taken between those points.



Conservative and nonconservative forces

- A **conservative force** allows conversion between kinetic and potential energy.
- Gravity and the spring force are conservative.
- The work done between two points by any conservative force
 - a) can be expressed in terms of a *potential energy function*.
 - b) is reversible.
 - c) is independent of the path between the two points.
 - d) is zero if the starting and ending points are the same.
 - e) is independent of its velocity or acceleration.

Conservative and nonconservative forces

- A force (such as friction) that is not conservative is called a **nonconservative force**, or a **dissipative force**.
- The work done by a nonconservative force
 - a) cannot be expressed in terms of a *potential energy function*.
 - b) is not reversible.
 - c) depends on the path between the two points.
 - d) is nonzero if the starting and ending points are the same.
 - e) may depend on its velocity or acceleration.

Nonconservative forces

- As an automobile tire flexes as it rolls, nonconservative internal friction forces act within the rubber.
- Mechanical energy is lost and converted to internal energy of the tire.
 - This causes the temperature and pressure of a tire to increase as it rolls.
 - That's why tire pressure is best checked before the car is driven, when the tire is cold.



Conservation of energy

- Nonconservative forces do not store potential energy, but they do change the *internal energy* of a system.
- **The law of conservation of energy** means that energy is never created or destroyed; it only changes form.
- This law can be expressed as

$$K_1 + U_1 = K_2 + U_2 + U_{\text{int},2}$$

$$\Delta K + \Delta U + \Delta U_{\text{int}} = 0.$$

Quiz

1. If the work done by a conservative force on a particle as it moves between two positions is $-10 \text{ N}\cdot\text{m}$, the change in its potential energy is _____.
A) $0 \text{ N}\cdot\text{m}$ B) $-10 \text{ N}\cdot\text{m}$
C) $+10 \text{ N}\cdot\text{m}$ D) None of the above.
2. Recall that the work of a spring is $W_{1-2} = -\frac{1}{2} k(s_2^2 - s_1^2)$ and can be either positive or negative. The potential energy of a spring is $U = \frac{1}{2} ks^2$. Its value is _____.
A) always negative B) either positive or negative
C) always positive D) an imaginary number!

Conservative force and potential energy

- In one dimension, a conservative force can be obtained from its potential energy function using:

Force from potential energy:

In **one-dimensional motion**,
the value of a conservative
force at point x ...

$$F_x(x) = -\frac{dU(x)}{dx}$$

... is the negative
of the derivative at x
of the associated
potential-energy function.

Conservative force and potential energy

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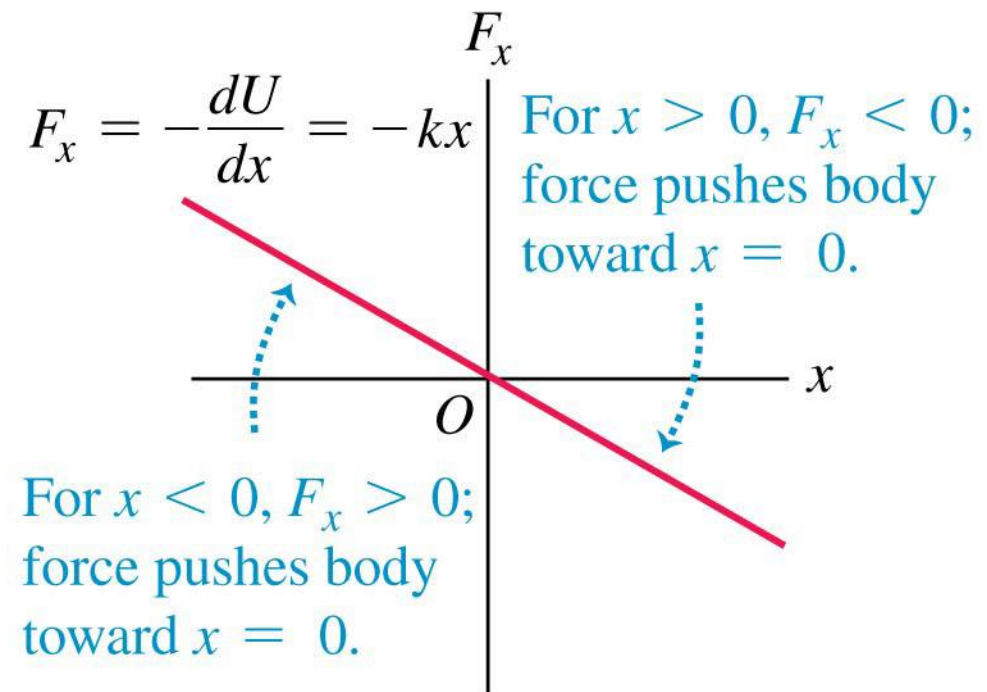
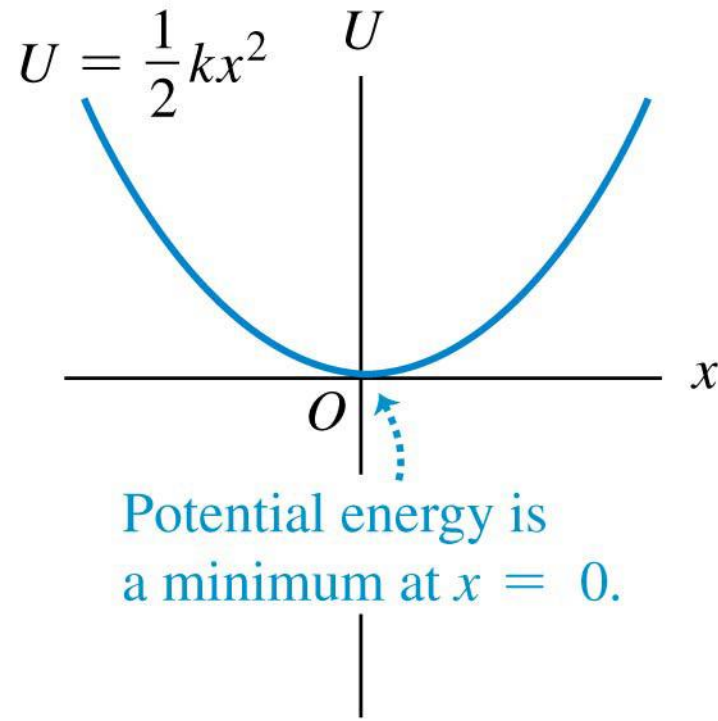
... is the negative of the derivative at x of the associated potential-energy function.

- In regions where $U(x)$ changes most rapidly with x , this corresponds to a large force magnitude.
- Also, when $F_x(x)$ is in the positive x -direction, $U(x)$ decreases with increasing x .
- A conservative force always acts to push the system toward *lower* potential energy.

Spring force and elastic potential

- Elastic potential energy and force as functions of x for an ideal spring.

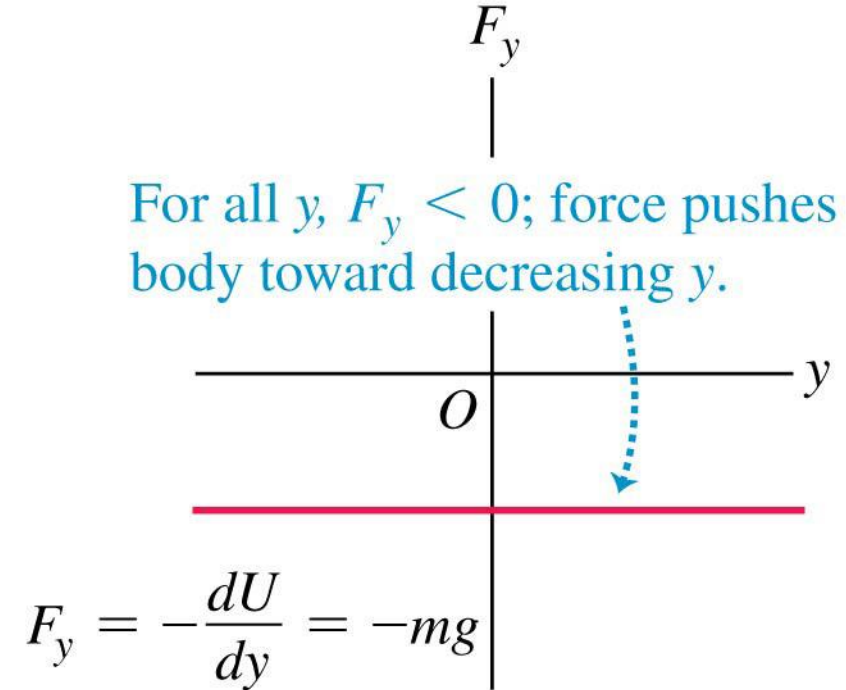
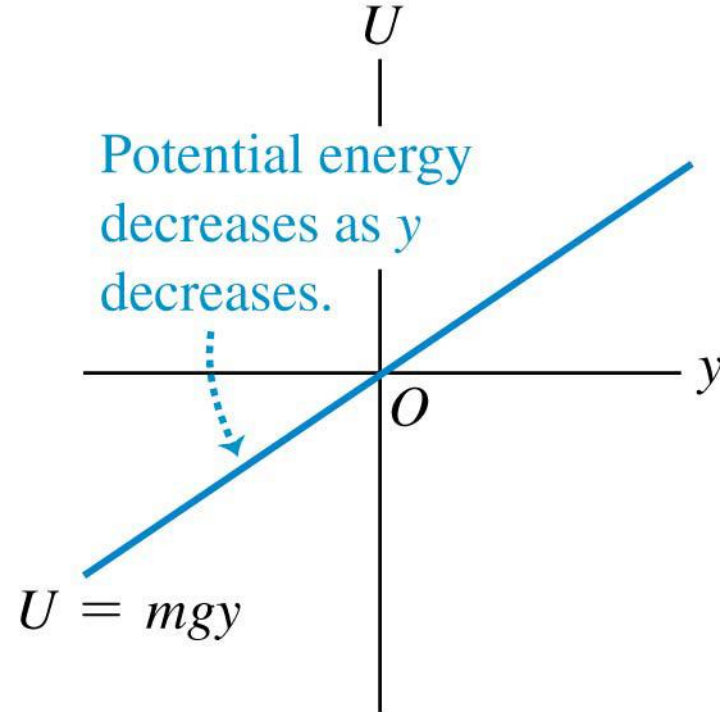
(a) Elastic potential energy and force as functions of x



Gravity and gravitational potential

- Gravitational potential energy and the gravitational force as functions of y .

(b) Gravitational potential energy and force as functions of y



The path-independence of the conservative force

- The path-independence is guaranteed by the fundamental theorem of calculus.

$$F_x(x) = -\frac{dU(x)}{dx}$$

$$\begin{aligned} W_{1-2} &= \int_{x_1}^{x_2} F_x(x) dx = - \int_{x_1}^{x_2} \frac{dU(x)}{dx} dx = - U(x) \Big|_{x_1}^{x_2} \\ &= U(x_1) - U(x_2) \end{aligned}$$

Force and potential energy in three dimensions

- In three dimensions, the components of a conservative force can be obtained from its potential energy function using partial derivatives:

Force from potential energy: In three-dimensional motion, the value at a given point of each component of a conservative force ...

$$F_x = -\frac{\partial U}{\partial x} \quad F_y = -\frac{\partial U}{\partial y} \quad F_z = -\frac{\partial U}{\partial z}$$

... is the negative of the partial derivative at that point of the associated potential-energy function.

Force and potential energy in three dimensions

- When we take the partial derivative of U with respect to each coordinate, multiply by the corresponding unit vector, and then take the vector sum, this is called the **gradient** of U :

Force from potential energy: The vector value of a conservative force at a given point ...

$$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right) = -\vec{\nabla}U$$

... is the negative of the gradient at that point of the associated potential-energy function.

Path-independence of the conservative force (3D)

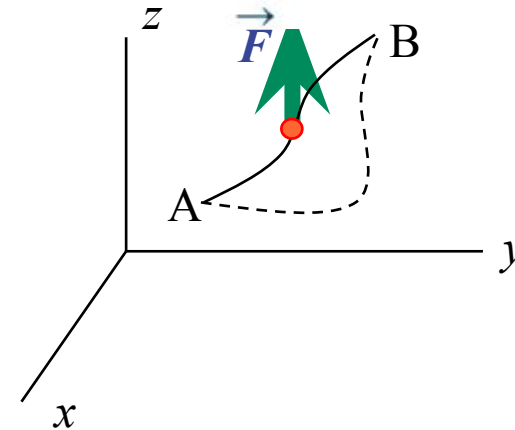
- To prove path independence,

$$\vec{F} = -\vec{\nabla}U$$

- we only need to prove

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{A \rightarrow B} \vec{F} \cdot d\vec{r} + \int_{B \rightarrow A} \vec{F} \cdot d\vec{r}$$



Proof

Gravitational potential (full expression)

Newton's gravity is given by $\mathbf{F} = - (G M m / r^2) \hat{\mathbf{r}}$.

Its potential is given by $U(r) = \int (G M m / r^2) dr = - G M m / r$.

Proof

Example

A body is projected from the surface of the Moon with speed u in any direction. Show that the body cannot escape from the Moon if $u^2 < 2GM/R$, where M and R are the mass and radius of the Moon.

Force and a graph of its potential-energy function

- For any graph of potential energy versus x , the corresponding force is $F_x = -dU/dx$.
- Whenever the slope of U is zero, the force there is zero, and this is a point of equilibrium.
- When U is at a minimum, the force near the minimum draws the object *closer* to the minimum, so it is a *restoring force*. This is called **stable equilibrium**.
- When U is at a maximum, the force near the maximum draws the object *away* from the maximum. This is called **unstable equilibrium**.

Unstable equilibrium

- Each of these acrobats is in unstable equilibrium.
- The gravitational potential energy is lower no matter which way an acrobat tips, so if she begins to fall she will keep on falling.
- Staying balanced requires the acrobats' constant attention.



Small oscillations

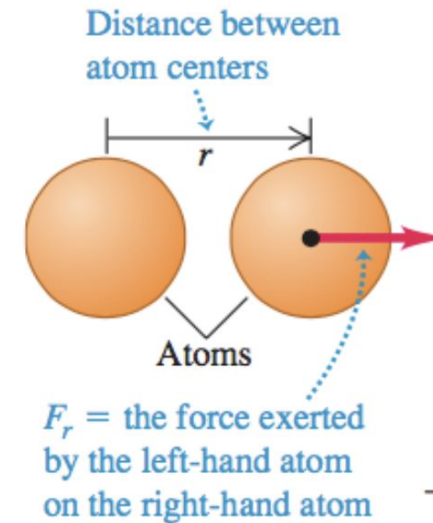
- At the minimum $x = d$, we can Taylor expand $U(x)$ by
$$U(x) = U(d) + (x - d)U'(d) + \frac{1}{2}(x - d)^2U''(d) + \dots$$
- The second term vanishes because it is a min.
- $F = -dU/dx = -(x - d)U''(d)$
- $m\ddot{x} + U''(d)(x - d) = 0$
- We got a SHM with $\omega = \sqrt{U''(d)/m}$

Example

- Two atoms having centers a distance r apart, with the equilibrium point at $r = R_0$. The potential is given by

$$U = U_0 \left[\left(\frac{R_0}{r} \right)^{12} - 2 \left(\frac{R_0}{r} \right)^6 \right]$$

What is angular frequency of the oscillation?



Summary

The gravitational potential energy (constant g)

The elastic potential energy

The mechanical energy conservation

The conservative force

The gravitational potential energy (full expression)

The stable and unstable equilibrium

The small oscillation near the minimum of the potential